Module -3
ASYMMETRIC CIPHERS

Principles of Public-Key cryptosystems

Public key cryptography has become an essential means of providing confidentiality, especially through its need of key distribution, where users seeking private connection exchange encryption keys. It also features digital signatures which enable users to sign keys to check their identities.

The approach of public key cryptography derivative from an attempt to attack two of the most complex problems related to symmetric encryption. The first issue is that key distribution. Key

Public key Cryptosystem − Asymmetric algorithms depends on one key for encryption and a distinct but related key for decryption. These algorithms have the following characteristics which are as follows −

It is computationally infeasible to decide the decryption key given only information of the cryptographic algorithm and the encryption key.

There are two related keys such as one can be used for encryption, with the other used for decryption.

- A public key encryption scheme has the following
- ingredients which are as follows −
- **Plaintext −** This is the readable message or information that is informer into the algorithm as input.
- **Encryption algorithm −** The encryption algorithm performs several conversion on the plaintext.
- **Public and Private keys −** This is a set of keys that have been selected so that if one can be used for encryption, and the other can be used for decryption.
- **Ciphertext −** This is scrambled message generated as output. It based on the plaintext and the key. For a given message, there are two specific keys will create two different ciphertexts.
- **Decryption Algorithm −** This algorithm get the ciphertext and the matching key and create the original plaintext.

- The keys generated in public key cryptography are too large including 512, 1024, 2048 and so on bits. These keys are not simply to learn. Thus, they are maintained in the devices including USB tokens or hardware security modules.

- The major issue in public key cryptosystems is that an attacker can masquerade as a legal user. It can substitutes the public key with a fake key in the public directory. Moreover, it can intercepts the connection or alters those keys.

- Public key cryptography plays an essential role in online payment services and ecommerce etc. These online services are ensure only when the authenticity of public key and signature of the user are ensure.

- The asymmetric cryptosystem should manage the security services including confidentiality, authentication, integrity and non-repudiation. The public key should support the security services including non-repudiation and authentication. The security services of confidentiality and integrity considered as an element of encryption process completed by private key of the user.

RSA Algorithm in Cryptography

- RSA(**R**ivest-**S**hamir-**A**dleman) Algorithm is an **asymmetric** or **public-key cryptography** algorithm which means it works on two different keys: **Public Key** and **Private Key**. The Public Key is used for **encryption** and is known to everyone, while the Private Key is used for **decryption** and must be kept secret by the receiver. RSA Algorithm is named after Ron **R**ivest, Adi **S**hamir and Leonard **A**dleman, who published the algorithm in 1977.

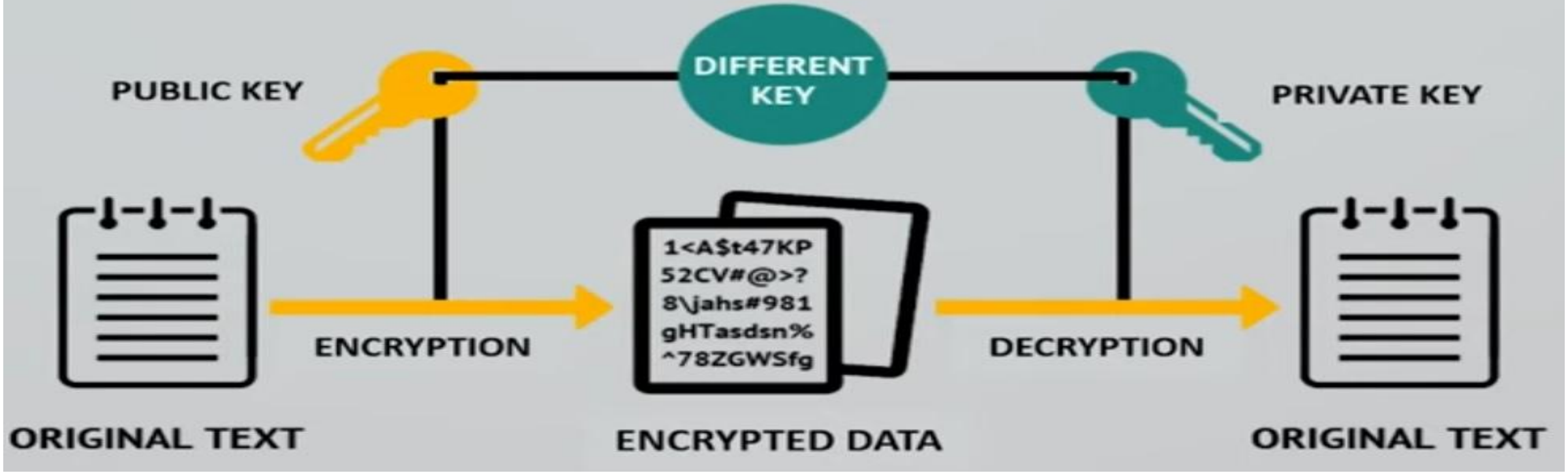
Example of Asymmetric Cryptography:

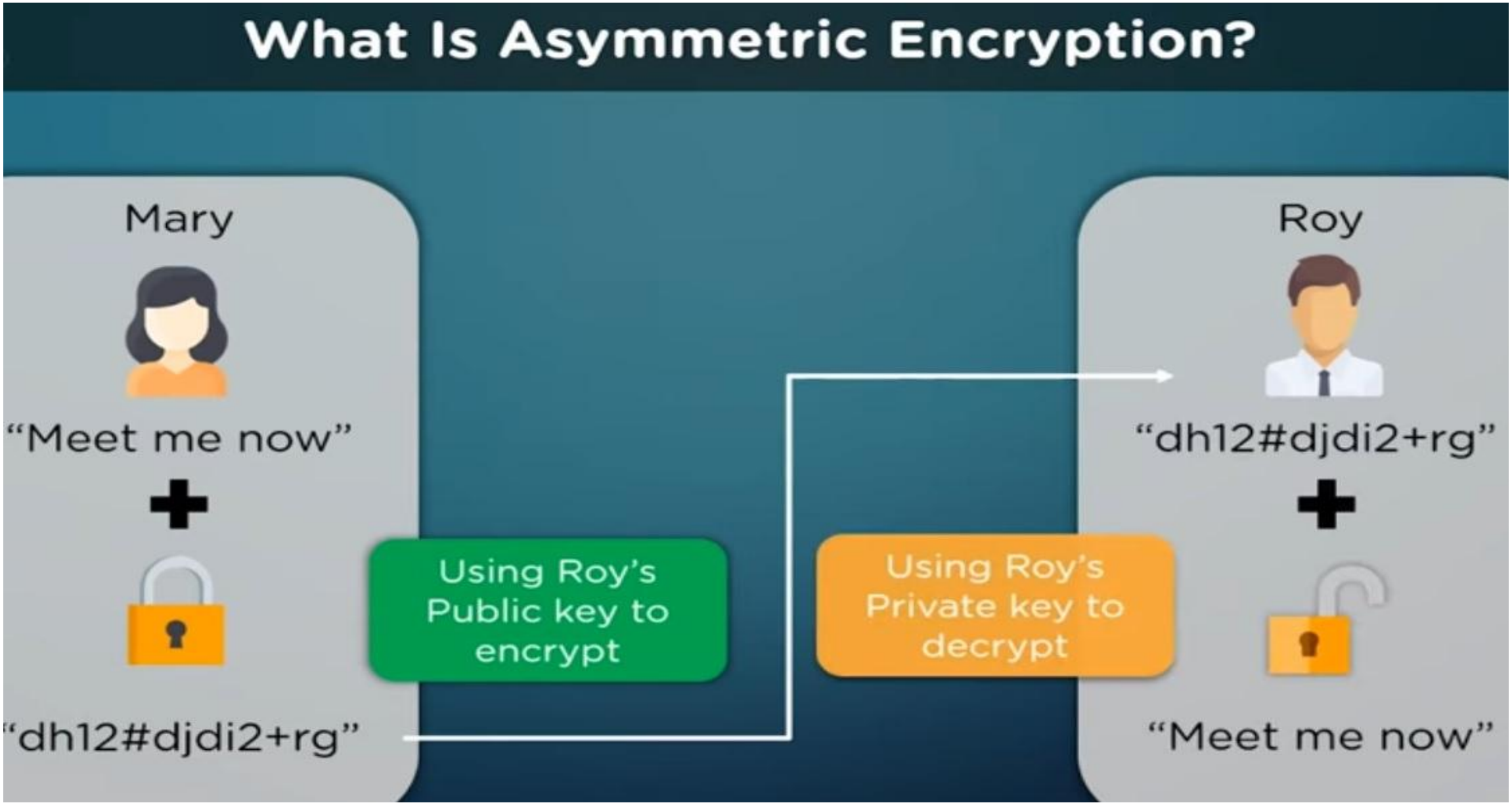
If Person **A** wants to send a message securely to Person **B**:

- Person A **encrypts** the message using Person B's **Public** Key.
- Person B **decrypts** the message using their **Private** Key.

What Is Asymmetric Encryption?

Two different keys are used in Asymmetric Encryption. Private key is used for encrypting the data, and public key is used for decrypting the data.




What Is Asymmetric Encryption?
Mary
"Meet me now"
"dh12#djdi2+rg"
Using Roy's Public key to encrypt
Using Roy's Private key to decrypt
Roy
"dh12#djdi2+rg"
"Meet me now"

Formula for Encryption and Decryption

- Encryption: C = P^e mod n
- Decryption: P = C^d mod n
- Public Key = { e, n }
- Private Key = { d, n }

RSA Key Generation

1) Consider two large prime numbers q, p.

2) Calculate $n = p \times q$.

3) $\phi(n) = (p - 1)(q - 1)$ [$\phi(n)$ Euler's Totient Function].

4) Choose a small number e, co-prime to $\phi(n)$, such that

$GCD(\phi(n), e) = 1$ and $1 < e < \phi(n)$.

5) Find d, such that $d \times e \bmod \phi(n) = 1$.

Example - RSA Key Generation

1) Two prime numbers: $p = 3$, $q = 5$.

2) Calculate $n = p \times q = 3 \times 5 = 15$.

3) Compute $\phi(n) = (p - 1)(q - 1)$:

$\phi(n) = (3 - 1) \times (5 - 1) = 8$.

4) Choose e such that $\gcd(e, \phi(n)) = 1$ and $1 < e < \phi(n)$.

Assume $e = 3$.

Possible values: $\gcd(3,8) = 1$,

$\gcd(5,8) = 1$,

$\gcd(7,8) = 1$

Finding d in RSA Key Generation

We need to find d such that:

- $d \times e \bmod \phi(n) = 1$

Given $e = 3$ and $\phi(n) = 8$:

- $d \times 3 \bmod 8 = 1$

Assume $d = 3$:

- $3 \times 3 \bmod 8 = 9 \bmod 8 = 1$

Thus, $d = 3$ is the correct value.

RSA Encryption and Decryption Example

Public Key: {e, n} = {3, 15}
Private Key: {d, n} = {3, 15}

Encryption:
Consider plaintext P = 8.
  C = P^e mod n
  C = 8^3 mod 15
  C = 512 mod 15
  C = 2

Decryption:
  P = C^d mod n
  P = 2^3 mod 15
  P = 8 mod 15
  P = 8 (retrieved plaintext)

Diffie-Hellman Key Exchange Algorithm

The Diffie-Hellman algorithm is being used to establish a shared secret that can be used for secret communications while exchanging data over a public network using the elliptic curve to generate points and get the secret key using the parameters.

- For the sake of simplicity and practical implementation of the algorithm, we will consider only 4 variables, one prime P and G (a primitive root of P) and two private values a and b.

- P and G are both publicly available numbers. Users (say Alice and Bob) pick private values a and b and they generate a key and exchange it publicly. The opposite person receives the key and that generates a secret key, after which they have the same secret key to encrypt.

Alice	Bob
Public Keys available = P, G	Public Keys available = P, G
Private Key Selected = a	Private Key Selected = b
Key generated = $x = G^a mod P$	Key generated = $y = G^b mod P$
Exchange of generated keys takes place	
Key received = y	key received = x
Generated Secret Key = $k_a = y^a mod P$	Generated Secret Key = $k_b = x^b mod P$
Algebraically, it can be shown that $k_a = k_b$	
Users now have a symmetric secret key to encrypt	

Diffie-Hellman Example

Step 1: **Alice** and **Bob** get public numbers **P = 23, G = 9**
Step 2: **Alice** selected a private key **a = 4** and
Bob selected a private key **b = 3**
Step 3: **Alice** and **Bob** compute public values
Alice: x =(9^4 mod 23) = (6561 mod 23) = **6**
Bob: y = (9^3 mod 23) = (729 mod 23) = **16**
Step 4: **Alice** and **Bob** exchange public numbers
Step 5: Alice receives public key **y =16** and
Bob receives public key **x = 6**
Step 6: Alice and Bob **compute symmetric keys**
Alice: ka = y^a mod p = 65536 mod 23 = 9
Bob: kb = x^b mod p = 216 mod 23 = 9
Step 7: 9 is the shared secret.

Elliptic Curve Cryptography

WHAT IS ECC?

- ECC is elliptic curve cryptography.
- The principal attraction of ECC, compared to RSA, is that it appears to offer equal security for a far smaller key size, thereby reducing processing overhead.

Elliptic curve

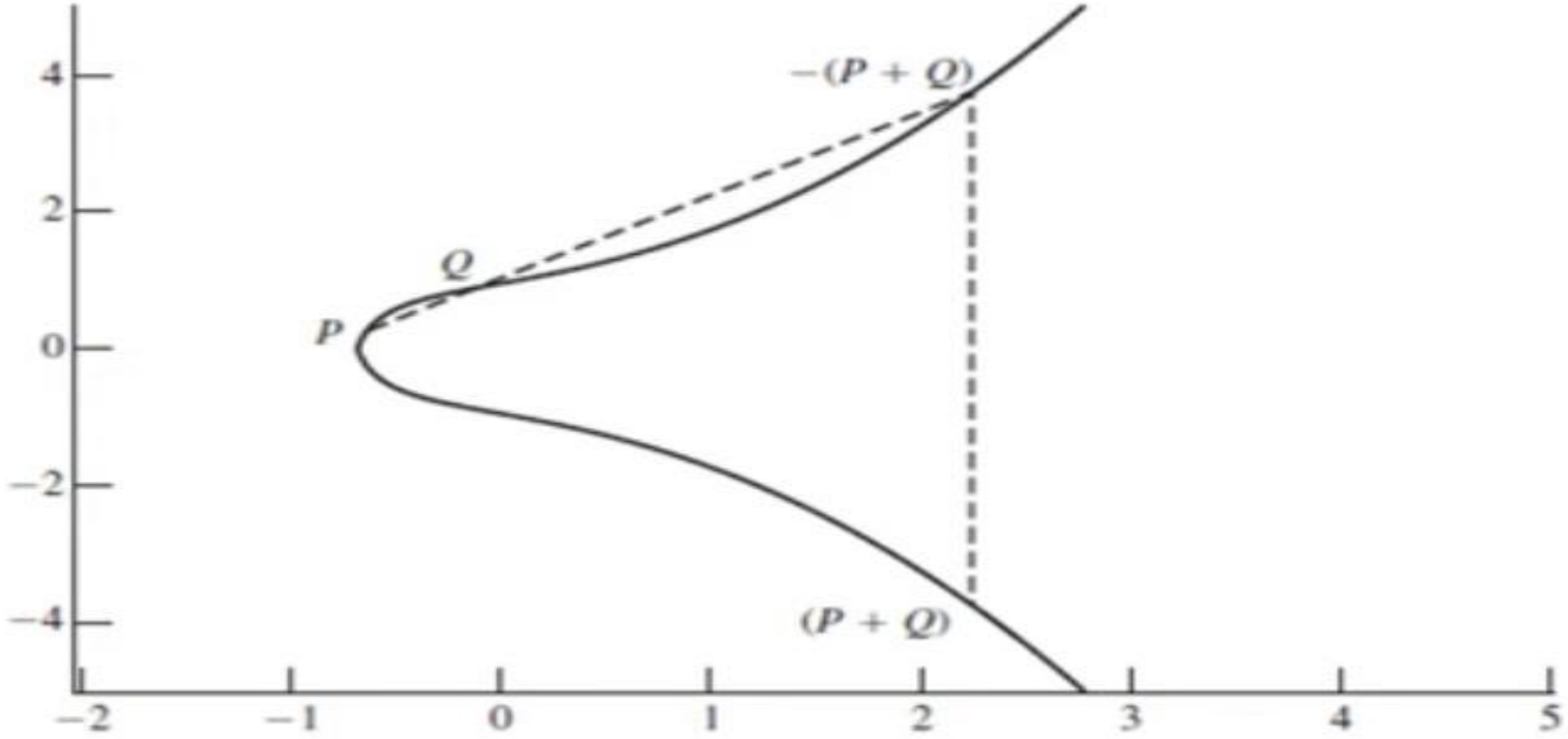
- An **elliptic curve** is defined by an equation in two variables with coefficients.

- For cryptography, the variables and coefficients are restricted to elements in a finite field, which results in the definition of a finite abelian group.

What is elliptic curve?

- Elliptic curves are not ellipses. They are so named because they are described by ***cubic equations***, similar to those used for calculating the circumference of an ellipse.
- $y^2 = x^3 + ax + b$

Elliptic curve - example



(b) $y^2 = x^3 + x + 1$

Parameters	ECC
Working algorithm	ECC is a cryptography technique that works just on a mathematical model of elliptic curves.
Bandwidth savings	ECC gives significant bandwidth savings over RSA.
Encryption process	The encryption process takes less time in ECC.
Decryption process	The decryption process takes more time.
Security	ECC is much safer than RSA and is currently in the process of adapting.

ECC & RSA Key Size Comparative Ratio Table

Category Type	ECC Key Size	RSA Key Size	Key Size Ratio
A	112	512	1:5
B	163	1024	1:6
C	192	1536	1:8
D	224	2048	1:9
E	256	3072	1:12
F	384	7680	1:20
G	512	15,360	1:30

How to find –P?

If $P = (x_P, y_P)$ Then $-P = (x_P, -y_P)$

For example:-

In $E_{23}(1,1)$, if $P = (13, 7)$, Then $-P = (13, -7)$.

$-7 \bmod 23 = 16$ i.e. $23 - 7 = 16$

So $-P = (13, -7) = (13, \underline{16})$.

Addition in elliptic curve arithmetic

If $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$ with $P \neq -Q$

Then $R = P + Q = (x_R, y_R)$

where

$$x_R = (\lambda^2 - x_P - x_Q) \bmod p$$

$$y_R = (\lambda (x_P - x_R) - y_P) \bmod p$$

λ Lambda

LAMBDA FORMULA

But to find λ :-

$$\lambda = \begin{cases} \left(\dfrac{y_Q - y_P}{x_Q - x_P}\right) \bmod p & \text{if } P \neq Q \\ \left(\dfrac{3x_P^2 + a}{2y_P}\right) \bmod p & \text{if } P = Q \end{cases}$$

EC Arithmetic Example

For example:-

$$P = (\underset{x_P}{3}, \underset{y_P}{10}), \quad Q = (\underset{x_Q}{9}, \underset{y_Q}{7}) \text{ in } E_{\underset{p}{23}}(\underset{a}{1}, 1) \quad \left[E_p(a, b)\right]$$

Find $P + Q$.

<u>Sol:</u>-

Here $P \neq Q$. So, $\lambda = \left(\dfrac{y_Q - y_P}{x_Q - x_P}\right) \bmod p.$

$$\lambda = \left(\frac{7 - 10}{9 - 3}\right) \bmod 23.$$

$$\Rightarrow \quad x = \left(\frac{-3}{6}\right) \bmod 23 = \left(-\frac{1}{2}\right) \bmod 23.$$

$$= -2^{-1} \bmod 23.$$

$$= -(-11) \bmod 23$$

$$= 11 \bmod 23$$

$$\boxed{x = 11}$$

To find $2^{-1} \bmod 23$.

$2^{-1} \bmod 23$

$\Rightarrow 23 = 2(11) + 1$

$\Rightarrow \cancel{23}^{0} + 2(-11) = 1 \bmod 23.$

$2(-11) = 1 \bmod 23$

So inverse of 2 is -11.

TO FIND X_R

$$x_R = (\lambda^2 - x_P - x_Q) \bmod p$$

$$= (11^2 - 3 - 9) \bmod 23 = (121 - 12) \bmod 23$$

$$= 109 \bmod 23.$$

$$\boxed{x_R = 17}$$

TO FIND Y_R

$$y_R = (\lambda(x_P - x_R) - y_P) \bmod p$$

$$= (11(3-17) - 10) \bmod 23$$

$$= (11(-14) - 10) \bmod 23$$

$$= (-154 - 10) \bmod 23$$

$$= -164 \bmod 23.$$

$$= 20 \bmod 23$$

$$\boxed{y_R = 20}$$

Hint

$$\boxed{\begin{aligned} 23 \times 7 &= 161 \\ 23 \times 8 &= 184 \\ 184 - 164 &= 20 \end{aligned}}$$

SOLUTION

- THE ANSWER IS

$$R = P + Q = (17, 20)$$